

CO5-AT1-Q1:

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Course Name: DAA for Complexity Analysis

Constraint-Based Game Strategy Solver (NP-Hard Problem)

Problem Understanding

Constraint-based game strategy problems require selecting the best possible moves while satisfying given game rules. Each game state represents a possible decision, and invalid moves must be eliminated.

The problem is considered NP-Hard because finding an optimal strategy requires exploring many possible combinations.

Algorithm Used

Backtracking

Constraints

- Valid moves only
- No repeated invalid states
- Game rules must be satisfied
- Losing paths must be avoided

Solution Approach

Steps:

1. Generate possible moves.
2. Check whether the move satisfies constraints.
3. Continue with valid choices.
4. Backtrack when a move leads to an invalid state.

Pruning Technique

Pruning removes unnecessary paths by checking constraints early.

Benefits:

- Reduces search space.
- Improves execution speed.

- Avoids exploring losing strategies.

Correctness

The algorithm explores all possible valid states. Since invalid choices are rejected immediately, the final solution satisfies all defined constraints.

Complexity Analysis

Worst Case Time Complexity:

$O(n!)$

Space Complexity:

$O(n)$

Conclusion

Backtracking provides an effective method for solving constraint-based problems. Although the search space is large, pruning techniques improve performance by eliminating unnecessary paths.